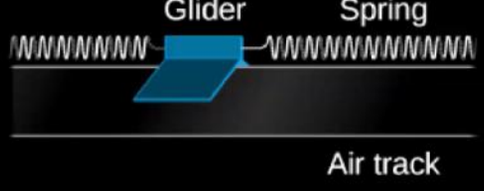
**EQUILIBRIUM:** A particle is said to be in equilibrium if net force on it is zero. Further if the particle is under influence of conservative forces alone then we know that the force acting on the particle is related to potential energy function as $\vec{F} = -\vec{\nabla}U = -\left[\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right]$, then if a particle is in equilibrium then each of the derivatives must be zero separately i.e for equilibrium:

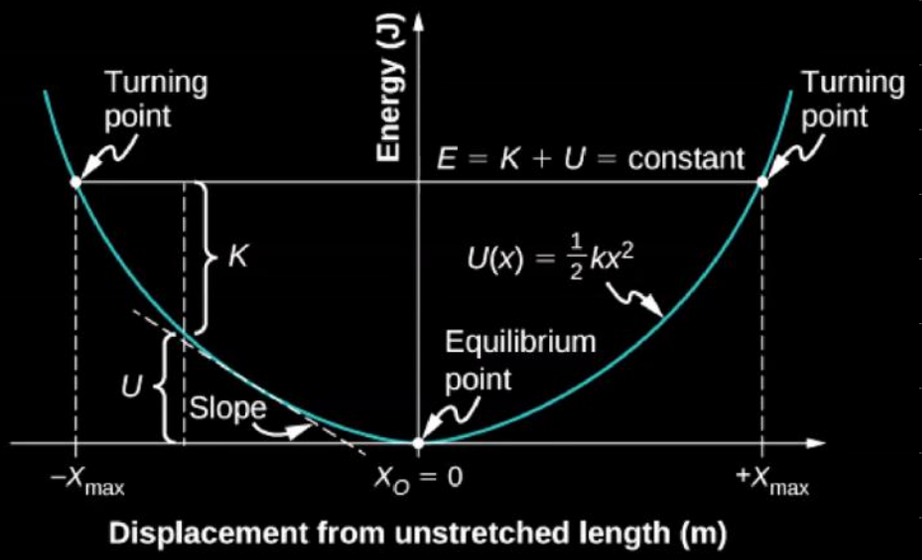
$$\frac{\partial U}{\partial x} = \frac{\partial U}{\partial y} = \frac{\partial U}{\partial z} = 0.$$




**DIRECTIONAL EQUILIBRIUM:** If all the three derivatives are not zero at some point while some of the derivatives are zero then we can say that particle is in equilibrium in that direction. For example if $\frac{\partial U}{\partial x} = 0$ but $\frac{\partial U}{\partial y}$ & $\frac{\partial U}{\partial z} \neq 0$ then we may say that there is equilibrium in $x$ direction by not in $y$ and $z$ directions.

**STABILITY AND EQUILIBRIUM (1D FORCE):**




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**STABLE EQUILIBRIUM:** If at a particular position of equilibrium a particle experiences a restoring force upon slight displacement, the equilibrium is said to be stable. Suppose a particle is subjected to a one dimensional conservative force $F(x)\hat{i}$. Then at position of equilibrium, $F(x) = -\frac{\partial U}{\partial x} = 0$. Now for stability,

upon increasing the $x$ coordinate from equilibrium position, a force must act in $-x$ direction and upon decreasing the $x$ coordinate, the force must act in $+x$ direction (i.e towards equilibrium position in both cases). Thus for stable equilibrium, $\frac{\partial F}{\partial x} < 0$ but $F_x = \frac{-\partial U}{\partial x}$ thus $-\frac{\partial^2 U}{\partial x^2} < 0$ or $\frac{\partial^2 U}{\partial x^2} > 0$

or we can simply say that for stable equilibrium, we should have a minimum of potential energy at the equilibrium position. Thus For stable equilibrium:

$$\boxed{\frac{\partial U}{\partial x} = 0} \quad \text{and} \quad \boxed{\frac{\partial^2 U}{\partial x^2} > 0}$$

→ Here since force is 1D, we actually mean usual derivative even though we have used partial derivative notation there. (In case of multidimensional force, these partial derivatives would imply directional stability).

**UNSTABLE EQUILIBRIUM:** If upon small displacement from equilibrium position, a particle experiences a force away from equilibrium position then the equilibrium is said to be unstable. Suppose a particle is subjected to $F_x(x)\hat{i}$ force then upon increasing the $x$ coordinate from equilibrium position, for unstability, the force must act in $+x$ direction and upon decreasing the $x$ coordinate, the force must act in $-x$ direction. Thus $\frac{\partial F}{\partial x} > 0$ but $U = -\frac{\partial F}{\partial x}$ therefore for unstable equilibrium,

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$\frac{\partial^2 U}{\partial x^2} < 0$, i.e the point must be a maximum of potential energy. Thus for unstable equilibrium:

$$\boxed{\frac{\partial U}{\partial x} = 0} \quad \text{and} \quad \boxed{\frac{\partial^2 U}{\partial x^2} < 0}$$

**NEUTRAL EQUILIBRIUM:** If a particle continues to experience zero force for slight displacements from the equilibrium position, the equilibrium is said to be neutral. Thus for neutral equilibrium, F vs x graph must be a horizontal line at zero level. Thus we can write, for neutral equilibrium:

$$\boxed{\frac{\partial U}{\partial x} = 0} \quad \text{and} \quad \boxed{\frac{\partial^2 U}{\partial x^2} = 0}$$

Note that $\frac{\partial^2 U}{\partial x^2} = 0$ does not guarantee a neutral equilibrium though it is a necessary condition. (Recall that when second derivative is zero, we call it a failure of second derivative test, and still the position can be a maximum or a minimum which can be inferred from higher order even derivatives).

**CALCULATING REVERSAL POINTS IN 1D CONSERVATIVE FIELD:**

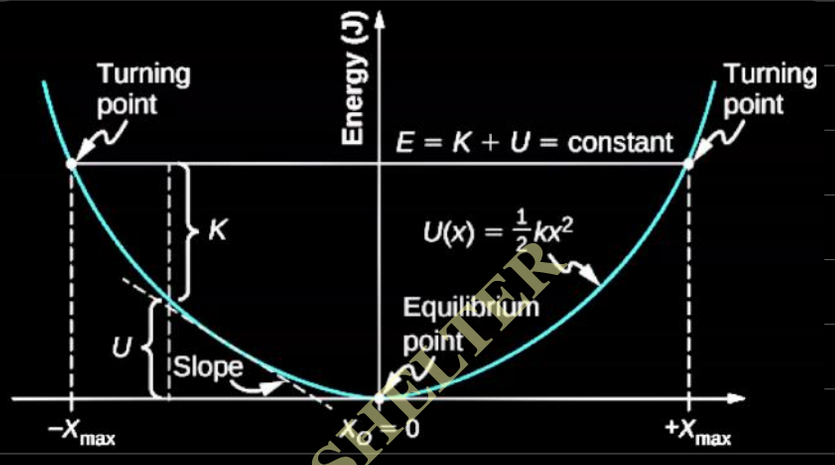
Suppose a particle is projected from $x = x_0$ with a speed $v = v_0$ along x direction where potential energy function is given as $U = U(x)$. Then total energy of the particle is:

$E = U(x_0) + \frac{1}{2} m v^2$. Since the force is conservative, total energy of the particle remains constant. As the particle travels, its PE changes and so also the KE to uphold the conservation law. Thus if somewhere along the direction of projection, the

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total energy becomes equal to potential Energy then particle has a momentary rest there. If that point is not a peak of potential energy, then the particle will reverse its direction, However if there exists no such point along the direction of projection, the particle will escape to ∞.

For instance consider the adjacent energy diagram. Here the points where the total Energy line intersects the potential energy curve are the points of reversal for a particle launched anywhere between $x_{max}$ and $x_{min}$.




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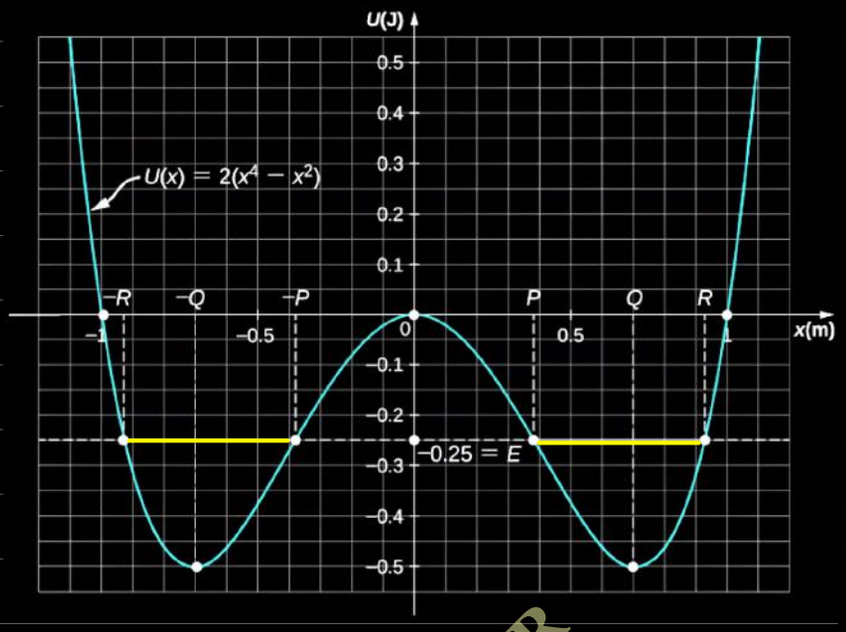
**EXAMPLE:** The potential energy for a particle undergoing one dimensional motion along x-axis is $U(x) = 2(x^4 - x^2)$ where U is in Joules and x is in metres. The particle is not subjected to any non-conservative forces and its total energy is constant at $E = -0.25$ J. (a) Is the motion of particle confined to any region on x axis, and if so what are they? (b) Are there any equilibrium points, and if so where are they and are they stable or unstable.

**Solution:** (a) We recognize that $U(\infty) = \infty > -0.25$ and also $U(-\infty) = \infty > -0.25$, thus particle can neither escape to ∞ nor to $-\infty$. Now in the regions where particle is bound, its KE must be greater than or equal to zero. Thus our required region(s) are given by $E - U(x) \geq 0$, or: $-\frac{1}{4} - 2(x^4 - x^2) \geq 0$

$$\Rightarrow -\frac{1}{4} - 2\left[\left(x^2 - \frac{1}{2}\right)^2 - \frac{1}{4}\right] \geq 0 \Rightarrow \left(x^2 - \frac{1}{2}\right)^2 \leq \frac{1}{8}$$

$$\Rightarrow \frac{1}{2} - \sqrt{\frac{1}{8}} \leq x^2 \leq \frac{1}{2} + \sqrt{\frac{1}{8}}$$. This gives us two allowed regions:

$x_p \leq x \leq x_R$ and $-x_R \leq x \leq -x_p$ (See adjacent figure), where $x_p = 0.38$ and $x_R = 0.92$.




(b) Now to find equilibrium points, we solve the equation:

$$\frac{dU}{dx} = 8x^3 - 4x = 0$$

which gives us $x = 0$ and $x = \pm x_0$ where $x_0 = 1/\sqrt{2} = 0.707$ m.

Now $\frac{d^2U}{dx^2} = 24x^2 - 4$. This is positive for $x = \pm \frac{1}{\sqrt{2}}$ and -ve for $x = 0$, thus $x = \pm \frac{1}{\sqrt{2}}$ are points of stable equilibrium and $x = 0$ is a point of unstable equilibrium and the particle is trapped in one of the regions marked by yellow line segments in the graph.

→ The particle will be oscillating in one of the two valley regions.

**EXAMPLE:** A particle moves under influence of a conservative force such that $U(x) = A/x^2 - B/x$ (a) Locate the point of equilibrium (b) Is it stable or unstable (c) What minimum KE must be given to the particle so that it may escape to infinity (From equilibrium.) (d) How close will it get to the origin if same speed were given towards the origin (e) For very small displacements around equilibrium, the restoring force may be approximate as $F \simeq -k\,\Delta x$ find the value of $k$. (A and B are positive constants).

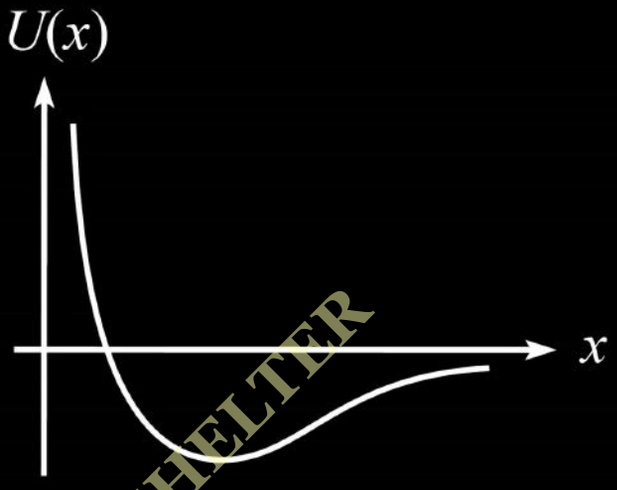
**Solution:** (a) $U'(x) = -2A/x^3 + B/x^2 = 0$ (For equilibrium)

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$\Rightarrow$ $x_0 = 2A/B$ (b) $U'' = 6A/x^4 - 2B/x^3$, putting $x = 2A/B$,

$U'' = B^4/8A^3 > 0$, $\therefore$ this must be a stable equilibrium.

(c) The potential energy graph of the particle looks as shown in adjacent figure. There are no peaks between the valley and $\infty$, so KE at the valley must be sufficient to account for gain in PE to reach infinity. So in limiting case:




$$KE + U(x_0) = U(\infty)$$

$$\Rightarrow KE_{min} = U_\infty - U(x_0)$$

$$= 0 - U(x_0)$$

Now $U(x_0) = U(2A/B) = \frac{-B^2}{4A}$, thus required min KE: $\frac{B^2}{4A}$ Ans.

(d) If the particle were given this much KE with velocity towards origin, it will come to a momentary rest where its PE becomes equal to total energy. In limiting case $E = 0$, thus for reversal point, $\frac{A}{x^2} - \frac{B}{x} = 0 \Rightarrow x = A/B$. This is the nearest distance of approach towards origin.

(e) By taylor approximation, $F(x + \delta x) = F(x) + \frac{dF}{dx}\delta x$, now $F(x_0) = 0 \Rightarrow$ required force $= -\frac{d^2U}{dx^2}\delta x = -\frac{B^4}{8A^3}\delta x = -k\delta x$

$\Rightarrow k = \frac{B^4}{8A^3}$ Ans.

**EQUILIBRIUM UNDER CENTRAL FORCES:** For a particle to remain stationary at a particular radius under central forces, by a logic similar to the one derived in a previous section:

$\boxed{\frac{dU}{dr} = 0}$ also for stable position $\boxed{\frac{d^2U}{dr^2} > 0}$,

and for unstable equilibrium, $\frac{d^2U}{dr^2} < 0$.

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**EXAMPLE:** The potential energy function for force between two atoms in a diatomic molecule is given by $U(r) = \frac{a}{r^{12}} - \frac{b}{r^6}$ where $r$ is the separation between atoms. What is the required bond dissociation energy assuming that atoms are initially at rest at equilibrium separation.

**Solution:** For equilibrium, $U'(r) = 0 \Rightarrow -12a/r^{13} + 6b/r^7 = 0$

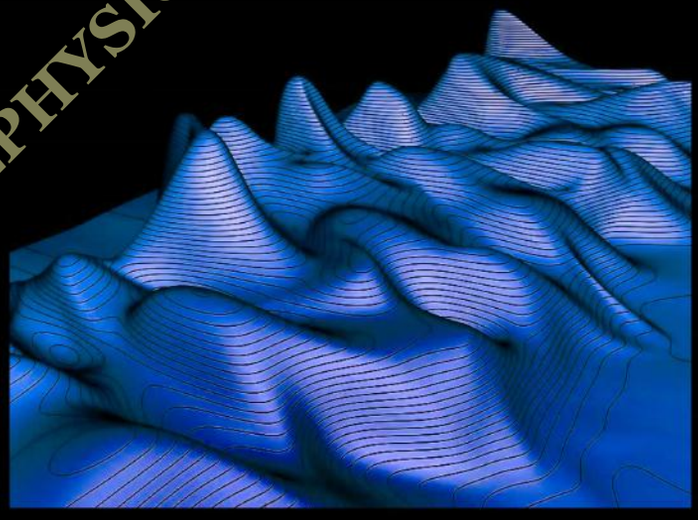
$\Rightarrow r_0 = \left(\frac{2a}{b}\right)^{1/6}$, let $K$ be the required KE for bond dissociation, then $K + U(r_0) = U(\infty)$ in limiting case.

$U(\infty) = 0, \Rightarrow K = -U(r_0) = -U\left((2a/b)^{1/6}\right) = b^2/4a$ Ans.

## EQUILIBRIUM IN 2D AND 3D FORCES:

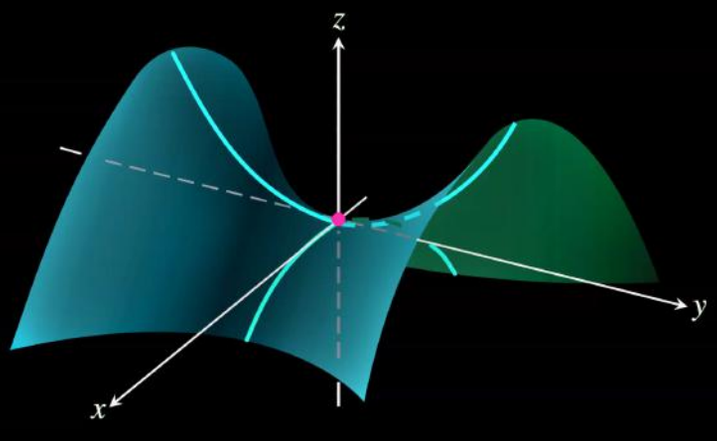
For a particle to be in complete stable equilibrium, its potential energy at that point must be lower than all points in its vicinity. Thus for a 2D field, the necessary condition for complete stable equilibrium is that,

$\frac{\partial U}{\partial x} = 0, \frac{\partial^2 U}{\partial x^2} > 0, \frac{\partial U}{\partial y} = 0$ & $\frac{\partial^2 U}{\partial y^2} > 0$



(This ensures a valley wrt $x$ and $y$ directions of travel, though it might still look like a hill from directions in between $x$ and $y$ directions).

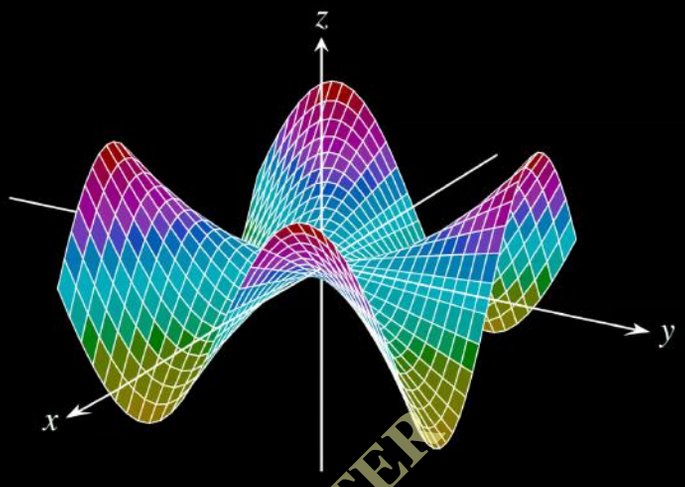
It is also possible to have surfaces which have directional stability in one direction and unstable in another direction. Such points are called saddle points or mountain passes. For instance in adjacent figure we have $\frac{\partial^2 z}{\partial y^2} > 0$ but




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$\frac{\partial^2 z}{\partial x^2} < 0$, thus it is y stable and x unstable.

→ Consider the adjacent surface. It looks like a peak from both x and y directions, But it looks like a valley when approached along bisectors of x and y axes. Thus $\frac{\partial^2 z}{\partial x^2} < 0$



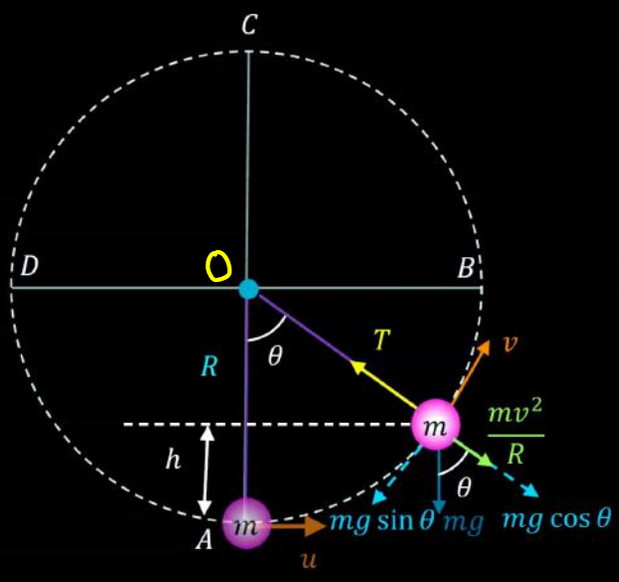

and $\frac{\partial^2 z}{\partial y^2} < 0$ does not guarantee unstability in every direction. Though it will be necessary condition for point to appear as a local maximum wrt 360° neighboring points.

→ For 3D forces, we can talk about equilibrium in terms of $\frac{\partial U}{\partial z}$ and $\frac{\partial^2 U}{\partial z^2}$ just as we did for 1D and 2D equilibria.

## Motion In A Vertical Circle:

### Vertical Circle With A String Bob System:

Consider a string Bob system as shown in the figure where Bob is imparted a speed u at the lowest point such that Bob is able to execute full vertical circle about the pivot O. We are interested in knowing the followings:




(a) Tension as a function of angle θ
(b) Minimum tension
(c) Minimum u to maintain the string taut throughout (or execute full circle)
(d) Speed at the topmost point corresponding to minimum u
(e) Value of u so as to perform oscillatory motion along arc BAD.

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(f) Can the string become lose anywhere on arc BAD for any value of $u$.

(g) What happens if $u$ is not sufficient to complete the full circle, but large enough to cross the horizontal level of the pivot.

Solution: (a) By Energy conservation: $v^2 - u^2 = 2gR(1-\cos\theta)$ ①

By force balance in centrifugal direction: $T = \frac{mv^2}{R} + mg\cos\theta$ ②

From ① & ②: $T = mg(3\cos\theta - 2) + \frac{mu^2}{R}$ ③

(b) Now as $\theta$ increases, Tension keeps decreasing for $\theta \le 180°$. Thus min tension occurs for $\theta = 180°$ and $T_{min} = \frac{mu^2}{R} - 5mg$ ④

(c) If string is to remain taut throughout, $T_{min} \ge 0$ thus,

$\boxed{u \ge \sqrt{5gR}}$ ← Min speed at bottom for full vertical circle.

(d) Now with limiting case, tension at topmost point is just zero, thus balancing forces in centrifugal direction, $mv^2/R = mg \Rightarrow$

$\boxed{v = \sqrt{gR}}$ ← Min speed at topmost point for full circle

(e) For oscillatory motion along Arc BAD, the velocity reversal must happen at B & D (i.e $v=0$ there). Thus $v^2 - u^2 = -2gH \Rightarrow$

$0 - u^2 = -2gR \Rightarrow \boxed{u = \sqrt{2gR}}$ ← Speed to oscillate in a semicircle.

→ If speed is smaller than this, oscillation happens in a smaller arc.

(f) String can never become lose anywhere in arc BAD because here both centrifugal force and gravity component act away from centre, thus tightening the string, However above the horizontal position, gravity component acts radially inwards and may cause loosening of the string.

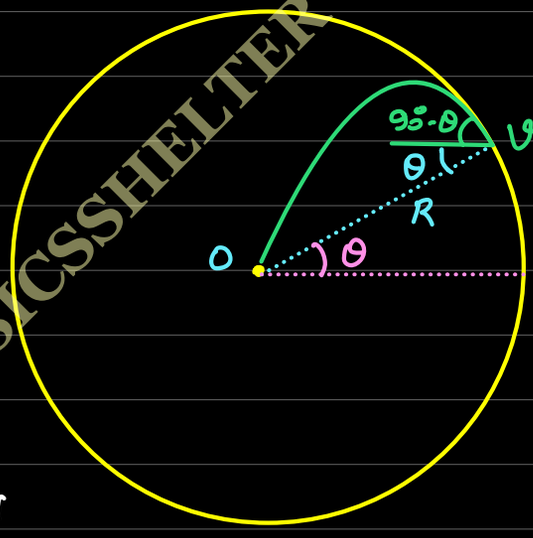
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(g) If $u$ is not large enough to complete the full circle i.e. $u < \sqrt{5gR}$ but sufficient to cross the pivot level i.e $u > \sqrt{2gR}$ ($\sqrt{2gR} < u < \sqrt{5gR}$) then the string will get lose somewhere above the horizontal level and projectile motion will start.

**EXAMPLE :** Find the launching speed of the string Bob system at the bottommost point so that the bob may hit the pivot.

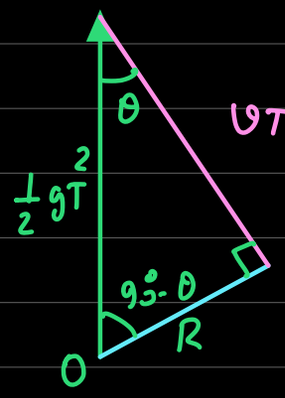
<u>Solution:</u> Let the string get loose at an angular position $\theta$ above the horizontal to meet our objective. Thus, by centrifugal force balance:

$$\frac{v^2}{R} = g\sin\theta \quad (1)$$




Now imagine a frame freely falling with $g$ with an initial velocity zero when the string got loose. Let $T$ be the time of hitting the pivot, then from such frame, we have a triangle diagram as shown. Now using sine law:

$$\frac{vT}{\cos\theta} = \frac{\frac{1}{2}gT^2}{1} = \frac{R}{\sin\theta} \quad (2)$$




From (2), $T = \frac{2v}{g\cos\theta}$, thus $\frac{vT}{\cos\theta} = \frac{2v^2}{g\cos^2\theta} = \frac{R}{\sin\theta}$

$\Rightarrow v^2 = \frac{gR\cos^2\theta}{2\sin\theta}$ (3) using (1) & (3), $gR\sin\theta = \frac{gR\cos^2\theta}{2\sin\theta} \Rightarrow \tan\theta = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin\theta = \frac{1}{\sqrt{3}}$, $\cos\theta = \sqrt{\frac{2}{3}}$, thus $v^2 = Rg\sin\theta = Rg/\sqrt{3}$

Also $u^2 - v^2 = 2gR(1+\sin\theta)$ (∵ Energy conservation), thus:

$$u^2 = v^2 + 2gR(1+\sin\theta) = Rg\sin\theta + 2gR(1+\sin\theta) = 2gR + 3gR\sin\theta$$

thus $u = \sqrt{(2+\sqrt{3})gR}$ Ans.

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**Think:** At what angle should the string get loose so as to again tighten at diametrically opposite point and what is corresponding launching speed?

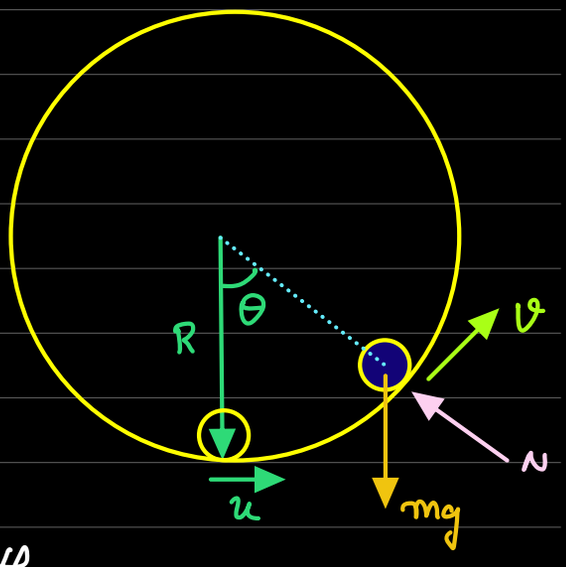
Solution: With new situation, the triangle diagram is still similar except that now $2R$ will replace $R$. Thus extreme RHS of equation 2 and RHS equation (3) also get $R$ replaced by $2R$ and we get $\tan\theta = 1$ $\Rightarrow \theta = 45°$, Now $u^2 = 2gR + 3gR\sin\theta = gR(2 + 3/\sqrt{2})$ accordingly,

$$u = \sqrt{gR(2+3/\sqrt{2})} \quad \text{Ans.}$$

→ If we wish to do the problem without frame change, we could use formula for range on inclined plane.

## Vertical Circle In A Frictionless Hoop:

Motion in a frictionless vertical hoop is essentially same as motion of a bob attached with a string, from dynamics point of view. This can be readily seen from energy and centrifugal force equations:

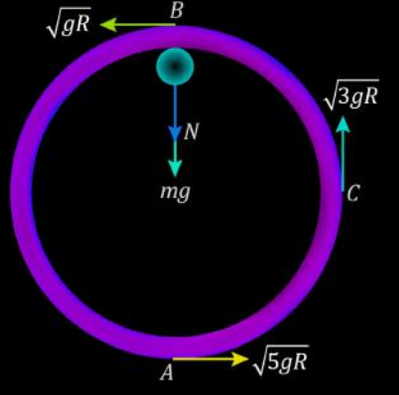



COME : $v^2 - u^2 = -2gR(1-\cos\theta)$ (1)

CEN : $N - mg\cos\theta = mv^2/R$ (2)

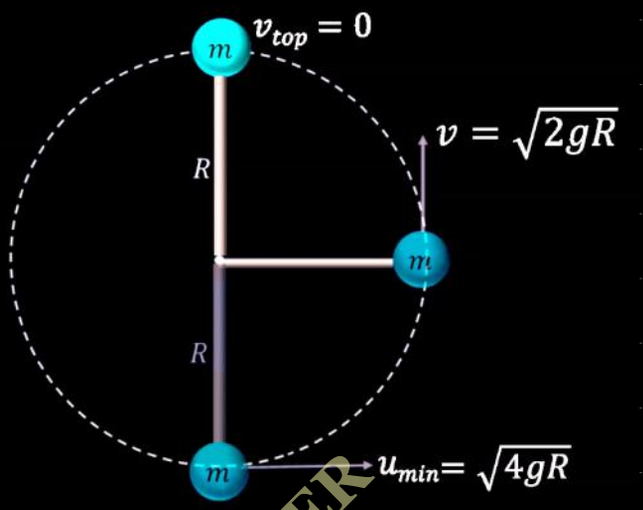
Notice that both equations are same as the ones we obtained for a string-bob arrangement except that here $N$ replaces $T$. Here if $\sqrt{2gR} < u < \sqrt{5gR}$, the normal reaction will become zero and projectile motion will start. For full circle, $u \geq \sqrt{5gR}$.

→ For the limiting case, the velocities at points A, B and C are as shown in the adjacent figure.




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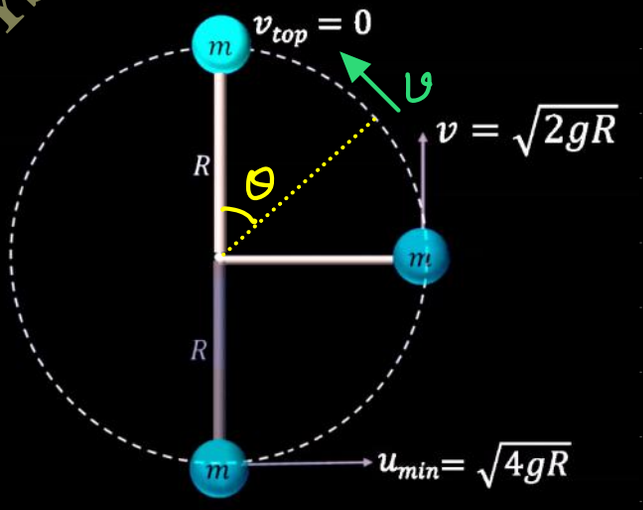
## Vertical Circle With A Massless Stick-Bob System:




Unlike the case of strings, a rod can also bear compressive loads, thus, a rod never gets "loose" unlike a strings. Now to find the critical velocity to complete the vertical circle we can just use energy conservation with velocity being 0 at the topmost point. Thus by energy conservation:

$$0^2 - u_{min}^2 = -2g \times 2R \Rightarrow \boxed{u_{min} = \sqrt{4gR}}$$

## Transition Angle For Load Becoming Tensile From Compressive:




As the motion progresses from bottom to top (critical velocity case), the force in the rod is initially tensile and it keeps reducing continuously until it becomes zero at some angle and thereafter, the force remains compressive until it reaches the top (downward motion is symmetric). We are interested in finding the transition angle $\theta$ where the force becomes zero. Now by energy conservation:

$$v^2 - 4gR = -2gR(1+\cos\theta) \quad (1)$$

By centrifugal equation: $\frac{v^2}{R} = g\cos\theta$ (2)

From (1) & (2), $Rg\cos\theta - 4gR = -2gR - 2gR\cos\theta \Rightarrow \cos\theta = 2/3$

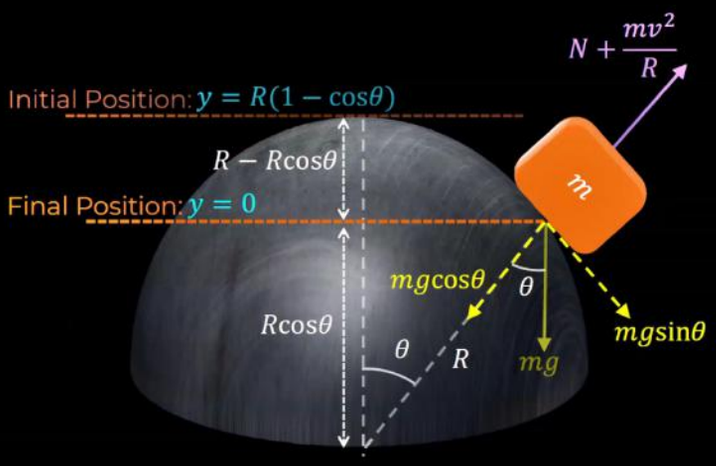
Thus transition angle:

$$\boxed{\theta_T = \cos^{-1}(2/3)}$$

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## Vertical Circular Arc On Smooth Hemisphere:

When a block is released from top of a smooth hemisphere, it moves along the surface till some angle and then loses contact at an angle $\cos^{-1}(2/3)$ from vertical. This can be easily shown using Energy and Centrifugal equations:



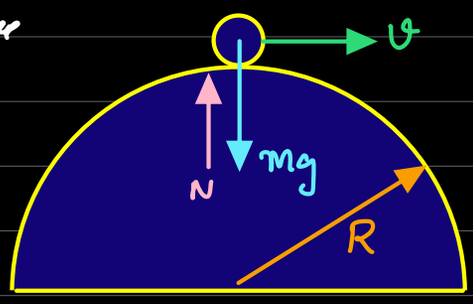

CEN: $v^2/R = g\cos\theta$ (1)

COME $v^2 - 0 = 2gR(1-\cos\theta)$ (2)

Solving (1) & (2), $\cos\theta = 2/3$. Thus we get the same answer as the transition angle with a shell case. (To continue circular motion beyond this point we will need an attractive force (or negative normal reaction, or "Tension").

## Flicking Velocity At Top Of Smooth Hemisphere For Immediate Contact Loss:

Suppose we flick a block on top of hemisphere with some velocity so that it continues on a circular arc for some time then, the centrifugal equation at the topmost point gives $N: mg - mv^2/R$, this means that as the flicking speed is increased, the normal reaction will continue to decrease, and for a certain speed, it will become zero. This speed is the minimum required speed for immediate loss of contact and is readily found by setting normal reaction to zero in centrifugal equation. Thus,

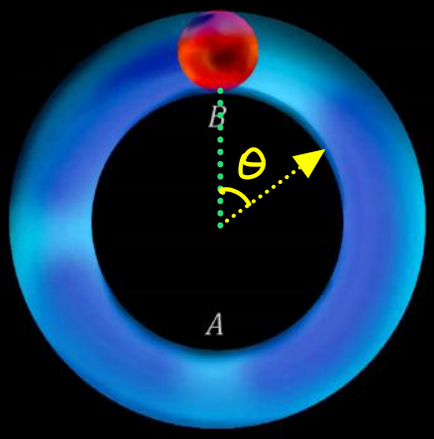



$m v_{min}^2/R - mg = 0 \Rightarrow$

$$\boxed{v_{min} = \sqrt{Rg}}$$

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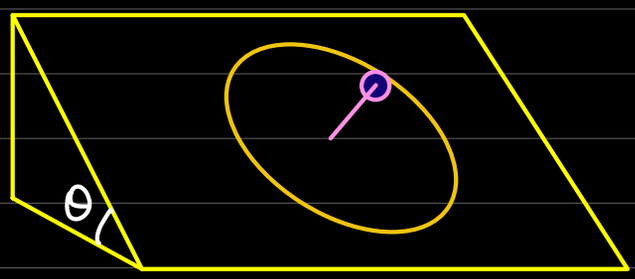
## Vertical Circle Inside A Tube:




From dynamics point of view, this problem is similar to vertical circle with a Bob and stick arrangement. Here the particle is constrained to go in a circle so minimum speed at bottommost point can be found by simply energy conservation (Assuming zero PE at topmost point). Thus $\boxed{u_{min} = \sqrt{4gR}}$

Here as the particles moves upwards, initially it presses against the outer wall and gradually the normal reaction reduces to zero at the transition angle $(\cos^{-1}(2/3))$ from the top after which it begins to press against the inner wall, the downward motion is symmetric to the upward motion.

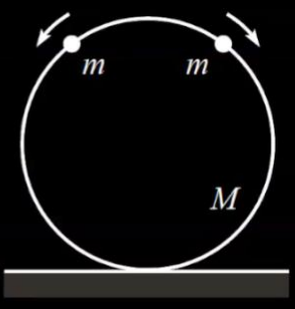
## Circular Motion On Inclined Plane:




All the cases of circular motion discussed so far can also be done on a smooth inclined plane however here, $g$ needs to be replaced by $g\sin\theta$ in all the results (i.e component of gravity in the plane of the incline), this is because $mg\cos\theta$ is always balanced by an equal and opposite normal reaction, and thus being balanced plays no role in the circular motion along the plane.

## Example: David Morin: 5.51:

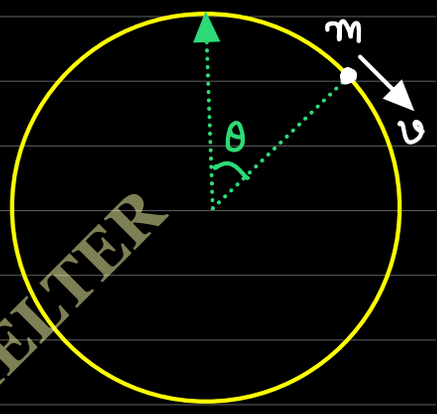
Two beads of mass $m$ are initially at rest at the top of a frictionless hoop of mass $M$ and radius $R$, which stands vertically on the ground. The beads are given tiny kicks, and they slide down the hoop, one to the right and one to the left, as shown in Fig. 5.37. What is the largest value of $m/M$ for which the hoop never rises up off the ground?




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**Solution:** As the beads slide down, up till an angle of $\cos^{-1}(R/3)$ from top, they experience an outward normal reaction and afterwards an inward normal, thus trying to pull the ring due to this reaction. The ring will experience least normal reaction from ground when vertical component of this pulling force is maximum.

Now consider a location where angular position is $\theta$.




$$v^2 = 2gR(1-\cos\theta) \quad (1) \qquad (\because \text{COME})$$

$$-N = mg\cos\theta - \frac{mv^2}{R} = mg\cos\theta - 2mg(1-\cos\theta)$$

$$= mg[3\cos\theta - 2)$$

Now we need to maximize vertical force $F_v = N\cos\theta$

$F_v = mg\cos\theta[2-3\cos\theta]$, so we must maximize $\cos\theta(2-3\cos\theta)$, let $\cos\theta = x$ then,

we must maximize $x[2-3x)$, $\therefore 2-6x = 0 \Rightarrow x = 1/3$

or $\boxed{\cos\theta = 1/3}$, at this instant vertical pulling force is $mg/3$, but there are two such beads, thus for limiting equilibrium,

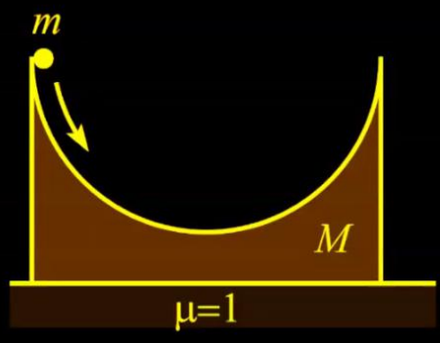
$$2mg/3 = Mg \Rightarrow \frac{m}{M} = 3/2 \text{ in limiting case.}$$

$\therefore$ our required answer is $m/M < 3/2$.

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## EXAMPLE: MORIN 5.52:

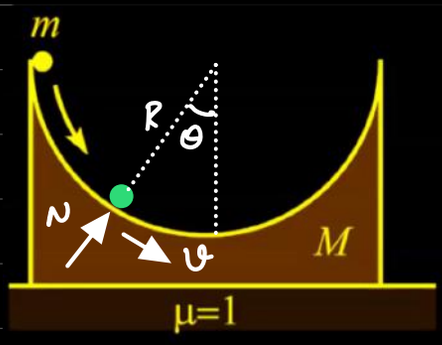
A hemispherical bowl of mass $M$ rests on a table. The inside surface of the bowl is frictionless, while the coefficient of friction between the bottom of the bowl and the table is $\mu = 1$. A particle of mass $m$ is released from rest at the top of the bowl and slides down into it, as shown in Fig. 5.38. What is the largest value of $m/M$ for which the bowl never slides on the table? *Hint*: The angle you're concerned with is *not* 45°.



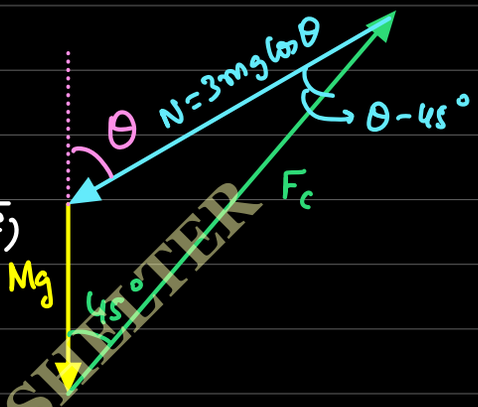

**Solution:** Considering the particle at a general angular position $\theta$, $v^2 = 2gR\cos\theta$ (1) $(\because$ COME$)$.

$$N = \frac{mv^2}{R} + mg\cos\theta \quad (2) \quad (\because \text{centrifugal})$$

$$\Rightarrow N = 3mg\cos\theta \quad (3)$$



Now the reaction of this N acts on the Bowl. We shall find the maximum allowable value of m so that bowl is in limiting equilibrium at this angular position. The bowl is in equilibrium under action of 3 forces namely N, Mg and contact force (which acts at 45° with vertical in limiting equilibrium since, $\mu = 1 = \tan 45°$). Using this force data, we construct the force triangle as shown:




Now using Sine law:

$$\frac{Mg}{\sin(\theta-45°)} = \frac{3mg\cos\theta}{\sin 45°} \Rightarrow m = \frac{M}{3}\frac{\sin 45°}{\cos\theta \sin(\theta-45°)}$$

Now since we do not want slipping for any value of $\theta$, we need to minimize m wrt $\theta$. Thus we need to maximize $\cos\theta \sin(\theta-45°)$

Differentiating and equating to zero:

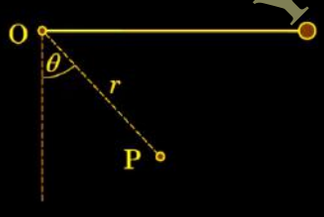
$$\cos\theta\cos(\theta-45°) - \sin\theta\sin(\theta-45°) = 0 \Rightarrow \cos(2\theta-45°) = 0 = \cos 90°$$

$$\Rightarrow \theta = 67.5° = 3\pi/8.$$ Thus largest value of m:

$$m_l = \frac{M}{3}\frac{\sin 45°}{\cos 67.5° \sin 22.5°} = \frac{M}{3}\frac{\sin 45°}{\sin^2(22.5°)} = \frac{2M}{3}\cot(22.5°)$$

$$m_l = \frac{2(\sqrt{2}+1)}{3} M \simeq 1.61\, M \text{ Ans.}$$

## EXAMPLE: PATHFINDER: BUILD: 24:




A small ball P is suspended from a fixed nail O with the help of a light inextensible cord of length $l$. There is another nail P at a distance $r$ from O. The ball is pulled aside stretching the cord horizontal and released. In the ensuing motion, the cord clings to the nail P and the ball begins to turn around nail P. Find range of values of $r$, so that the ball makes a complete circular turn around the nail P. Ignore air resistance and radii of the nails.

Solution: For vertical circle (limiting case) speed at the bottomost point is $u = \sqrt{5gR}$ and at an angle $\theta$ from vertical:

$$v^2 = u^2 - 2gR(1-\cos\theta) = gR[3+2\cos\theta]$$, in general,

$$v^2 > gR[3+2\cos\theta)$$

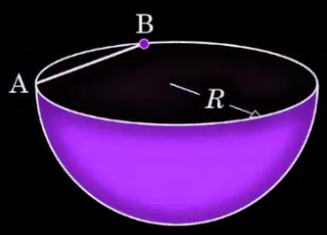
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For our case, $v^2 = 2gl\cos\theta$ ($\because$ COME), $R = l - x$ thus:

$$2gl\cos\theta > g(l-x)[3+2\cos\theta] \Rightarrow \frac{l-x}{l} < \frac{2\cos\theta}{3+2\cos\theta}$$

$$\Rightarrow x > l\left[1 - \frac{2\cos\theta}{3+2\cos\theta}\right] = \frac{3}{3+2\cos\theta}\, l \quad \text{Ans.}$$

## EXAMPLE: PATHFINDER: CHECK 16:




Consider a fixed hemispherical bowl of radius $R$ with its opening horizontal. One end of a light rod of length $l = R$ is hinged at a point A on the brim of the bowl. A small bead of mass $m$ is attached at the other end the rod. Initially the bead is held at a point B also on the brim and then released. Find the tensile force in the rod when the bead passes the lowest position. Acceleration of free fall is $g$.

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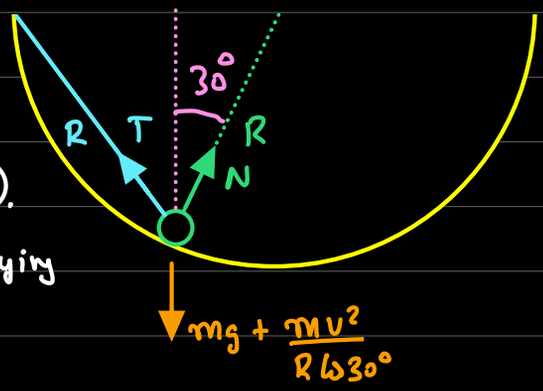
**Solution:** Here the locus of the particle is on a circle formed by intersection of the hemisphere with a sphere of radius $R$ centred at A ($\because$ distance from A is constant).

Now by energy conservation:

$$v^2 = 2gR\cos 30^\circ$$

$$\Rightarrow \text{Centrifugal force} = \frac{mv^2}{R\cos 30^\circ} = 2mg$$

($\because$ Particle is moving in a circle of radius $R\cos 30^\circ$).



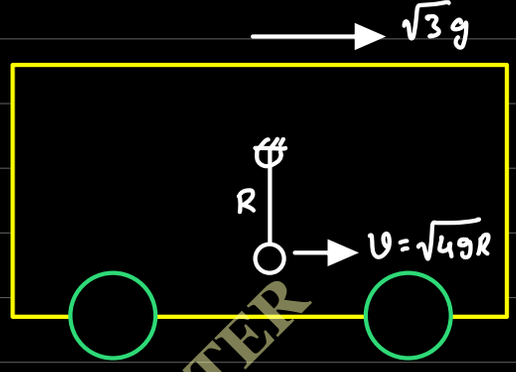

Now clubbing centrifugal force with $mg$ and applying Lami's theorem:

$$\frac{3mg}{\sin 60^\circ} = \frac{T}{\sin 150^\circ} = \frac{N}{\sin 150^\circ}$$

$$\Rightarrow T = 3mg \cdot \frac{\sin 150^\circ}{\sin 60^\circ} = 3mg\tan 30^\circ = \sqrt{3}\,mg \quad \text{Ans.}$$

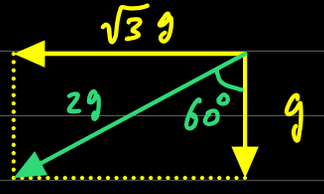
**VERTICAL CIRCLE IN ACCELERATED VEHICLES:** All standard formulae for standard cases can be applied to translationally accelerated vehicles, just that we need to replace $\vec{g}$ with effective gravity i.e $\vec{g}_{eff} = \vec{g} - \vec{a}$ ($\because m\vec{g}_{eff} = m\vec{g} - m\vec{a}$) where $\vec{a}$ is

the acceleration of the vehicle. However, one must interpret the words "topmost point" and "bottommost point" carefully. For our standard formulae, these words would imply maximum and minimum coordinates along effective gravity and not earth's gravity.

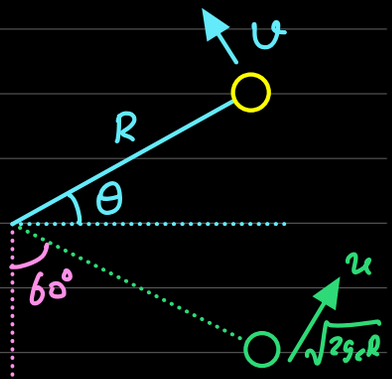



**EXAMPLE:** A bob is hanging in a vehicle, pivoted from a massless string of length R. At the bottommost point, the bob is given a speed $\sqrt{4gR}$ and immediately afterwards vehicle begins to move with an acceleration $\sqrt{3}g$. At what angle below horizontal does the string become slack.

**Solution:** Here effective gravity acts at an angle of 60° from the vertical thus for ease of visualization we can see the particle launched with a speed $\sqrt{2g_eR}$ from 60° position.




Now drawing the effective gravity vertically,




Now let string become loose at angle θ above horizontal, then:

$$\frac{v^2}{R} = g_e \sin\theta \quad (1) \quad (\because \text{centrifugal balance}).$$

also: $v^2 = u^2 - 2g_eR[\sin\theta + \sin 30°]$

$= 2g_eR - 2g_eR[\sin\theta + 1/2]$

$\Rightarrow v^2 = g_eR[1 - 2\sin\theta]$ (2) Now from (1) & (2):

$$Rg_e\sin\theta = Rg_e[1-2\sin\theta] \Rightarrow 3\sin\theta = 1 \text{ or } \sin\theta = 1/3 \Rightarrow \theta \approx 19.5°$$

i.e 19.5° above horizontal as defined using effective gravity, Now we need to convert it in terms of "horizontal" as seen from earth, Thus we need to subtract 60° from this. Thus string becomes loose at an angle (19.5° − 60°) above horizontal i.e 40.5° below Horizontal Ans.

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→ For a rough idea without using a calculator we could use $\sin\theta \approx \theta$ i.e $\theta \approx 1/3$ rad $= 60°/\pi \approx 19.1°$ (which is very close to 19.5°).

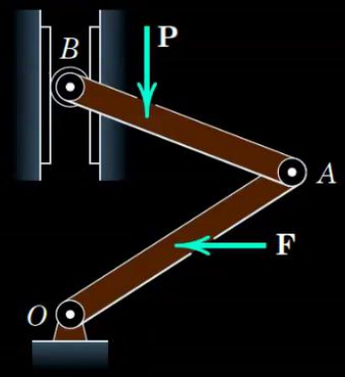
→ $\sin\theta \approx \theta - \theta^3/6$ is fairly accurate till 60° (less than 1% error) (at 90° error is about 8%).

## METHOD OF VIRTUAL WORK: (STATIC EQUILIBRIUM): (OPTIONAL)

### THREE TYPES OF FORCES IN MECHANICAL SYSTEMS:

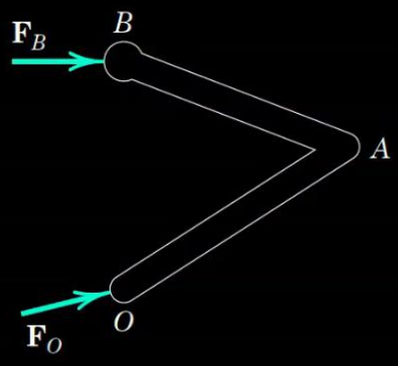
For the purpose of applying method of virtual work for solving equilibrium problems, we need to classify the forces based on their capability or incapability to do work for small perturbation of a mechanical system from its equilibrium position. We can classify the forces in to three types:

**ACTIVE FORCES:** These are external forces acting on a system which are capable of doing work upon small perturbation of the system from equilibrium configuration. For instance, in the figure P and F are active forces which will do work if B were to shift upwards or downwards.



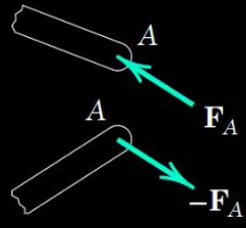
(a) Active forces

**REACTIVE FORCES:** These are the forces which act at the supports where either displacement is zero or perpendicular to the force upon slight perturbation of the system within its constraints. For instance in the figure, $F_O$ and $F_B$ are reactive forces.



(b) Reactive forces

**INTERNAL FORCES:** These are the forces in the connections between members of a mechanical system. If there is no kinetic friction between members, the sum of work done by all internal force pairs in a mechanical system is zero (Recall the section where we discussed zero work pairs).



(c) Internal forces

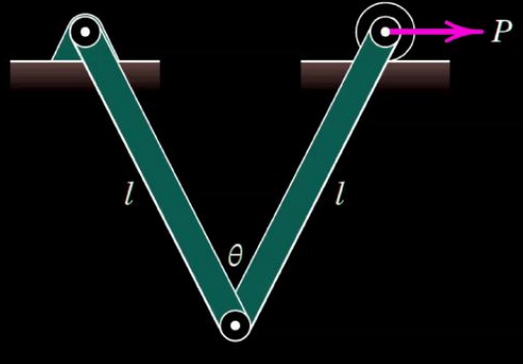
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**PRINCIPLE OF VIRTUAL WORK:** Recall that for any system to be in equilibrium under action of conservative forces, the derivative of potential energy wrt small perturbation of the system must be zero (The perturbation could be expressed either in terms of angular coordinate $\theta$ or linear coordinate). Thus for a system in equilibrium, $dU/d\theta = 0$ or $dU/dx$ is zero where $x$ and $\theta$ are suitable variables defining perturbation from equilibrium position. The equilibrium configuration only depends on the forces and not on the sources of the forces. Thus active forces can always be imagined to be applied using compressed or stretched springs. Now since $dU/dx = 0$ this means that for small change in perturbation variable $x$, $dU = 0$ which also means that for slight change in $x$, sum of work done by all active forces must be zero. Such a work is known as virtual work (we are imagining a perturbation while maintaining the active forces). Thus we can state the virtual work principle as:

The virtual work done by external active forces on an ideal mechanical system (made of frictionless rigid links) in equilibrium is zero for any and all virtual displacements consistent with constraints.

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**EXAMPLE (MERIAM KRAIGE):**



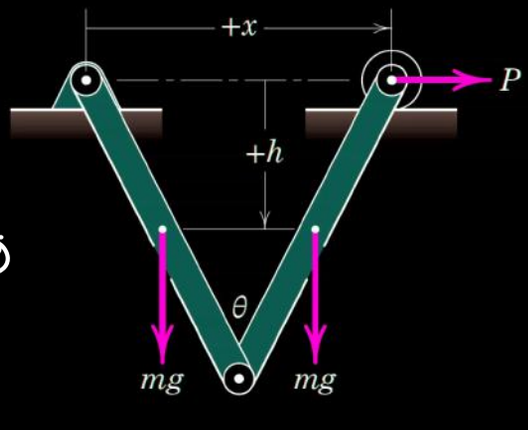

Each of the two rods has a mass $m$ and length $l$ and is supported and loaded as shown in the figure. Find angle $\theta$ for equilibrium.

Solution: Consider the adjacent figure, showing the coordinates of CM and loading point of P. Now by virtual work:

$$mg \times \delta h \times 2 + P \times \delta x = 0 \quad (1)$$

Now $h = \frac{1}{2} l \cos(\theta/2) \Rightarrow \delta h = -\frac{1}{4} l \sin(\theta/2)\, \delta\theta$ (2)

$$x = 2l \sin(\theta/2) \Rightarrow \delta x = l \cos(\theta/2)\, \delta\theta \quad (3)$$



using these in equation ①:

$$Pl\cos(\theta/2)\,\delta\theta - 2mg\frac{l}{4}\sin\left(\frac{\theta}{2}\right)\delta\theta = 0$$

$$\Rightarrow \tan(\theta/2) = 2P/mg \Rightarrow \theta = 2\tan^{-1}(2P/mg) \text{ Ans.}$$

Think: Can you do this problem by torque balance?

Solution: Considering the combined FBD of the system, we can see that problem has a perfect left right symmetry. Thus the pin joint connecting the two rods must offer equal and opposite horizontal (left & right) reactions to the two rods which must each be equal to P.

This gives us the force diagram as shown in the next figure.

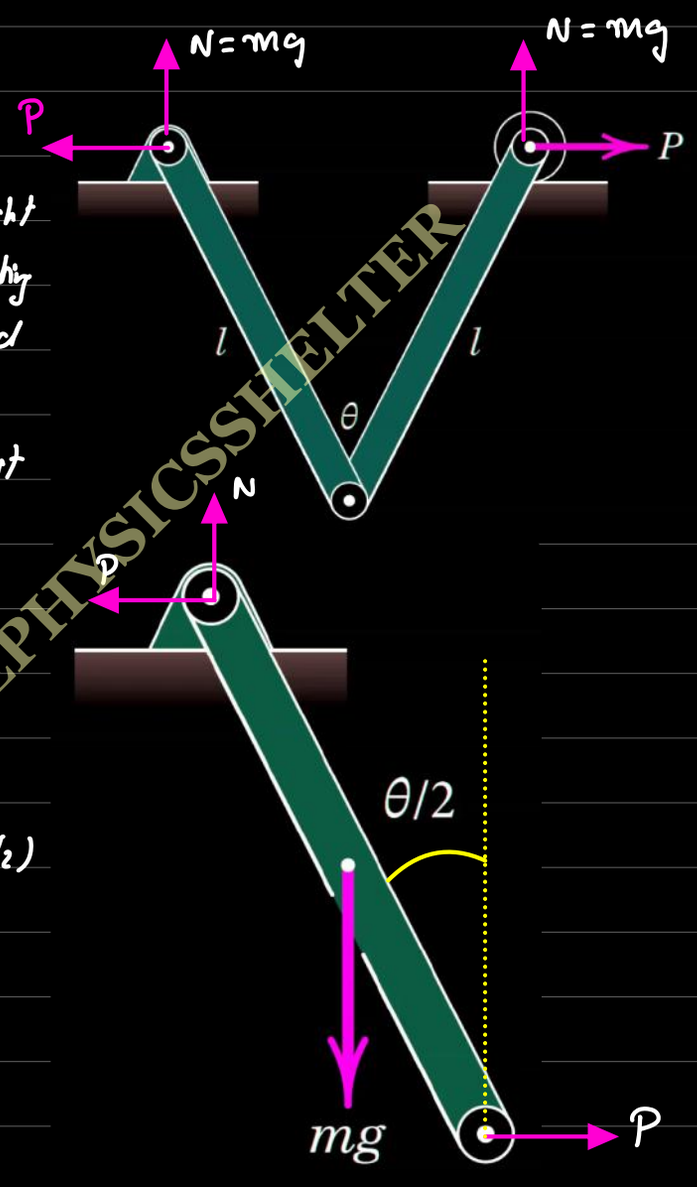
Now balancing torques about the fixed pin,

$$P \times l\cos(\theta/2) = mg\frac{l}{2}\sin(\theta/2)$$

$$\Rightarrow \tan(\theta/2) = 2P/mg.$$

$$\Rightarrow \theta = 2\tan^{-1}(2P/mg)$$

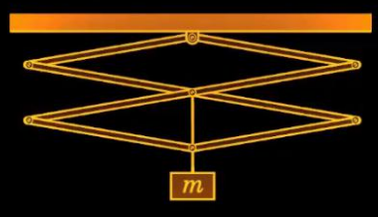
which is the same result that we obtained using method of virtual work.



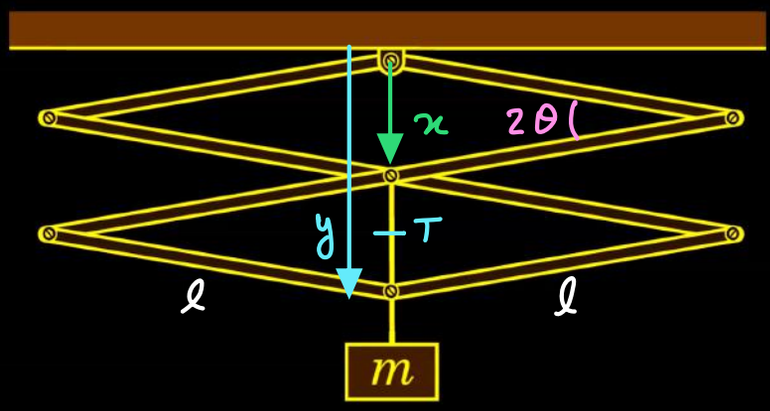

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EXAMPLE: PATHFINDER: BUILD 31:

Two long and four half as long light rods are hinged at their ends to form a pantograph consisting of two identical rhombi. The pantograph is suspended from the ceiling, the lowest hinge of the pantograph is connected to the hinge above it by a light inextensible cord and a load of mass $m$ is suspended from the lowest hinge as shown in the figure. Find the tensile force developed in the cord connecting the two hinges.



**Solution:** Consider the adjacent diagram showing the coordinates of points where active forces will be doing the work. (Here active forces are string tension and weight). Thus:




$$T\delta x - T\delta y + mg\,\delta y = 0$$

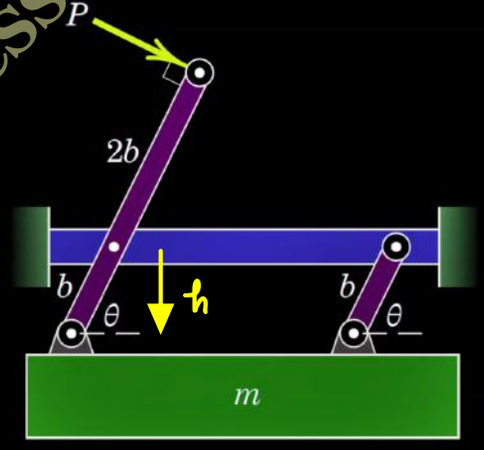
Now $x = 2\ell\sin\theta \Rightarrow \delta x = 2\ell\cos\theta\,\delta\theta$

$y = 4\ell\sin\theta \Rightarrow \delta y = 4\ell\cos\theta\,\delta\theta$ thus:

$$T \times 2\ell\cos\theta\,\delta\theta - T \times 4\ell\cos\theta\,\delta\theta - mg \times 4\ell\cos\theta\,\delta\theta = 0 \Rightarrow T = 2mg \quad \text{Ans.}$$

**EXAMPLE (MERIAM KRAIGE):**

For a given force P, determine the angle θ for equilibrium. Neglect the mass of the links.




**Solution:** The depth h is given by,

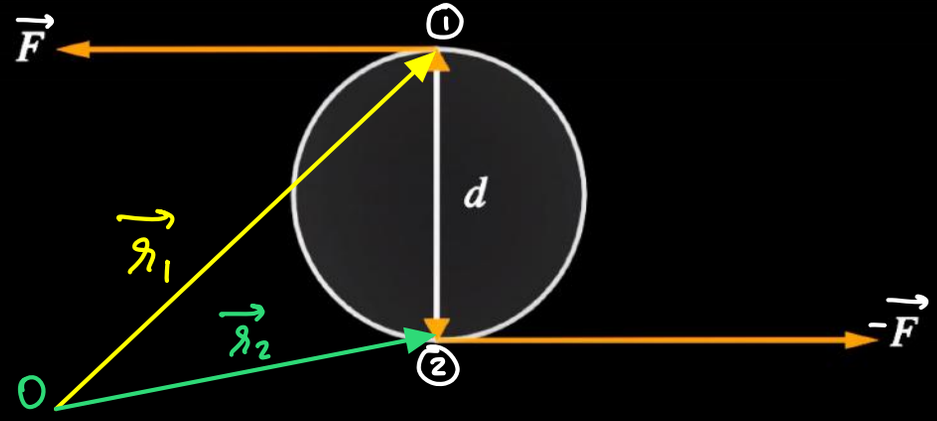
$h = b\sin\theta$ thus $\delta h = b\cos\theta\,\delta\theta$

Now if θ changes to θ+δθ, total work done by active forces: $mg\,b\cos\theta\,\delta\theta - P \times (2b\,\delta\theta) = 0$

$\Rightarrow \cos\theta = 2P/mg \Rightarrow \theta = \cos^{-1}(2P/mg)$ Ans.

**COUPLE:**

Two equal and opposite forces acting on a rigid body form a couple. The net torque of a couple is independent of point about which torque is being considered.




This can be easily seen as follows. Consider the forces $\vec{F}$ and $-\vec{F}$ acting on a disc having position vectors $\vec{r}_1$ and $\vec{r}_2$ wrt

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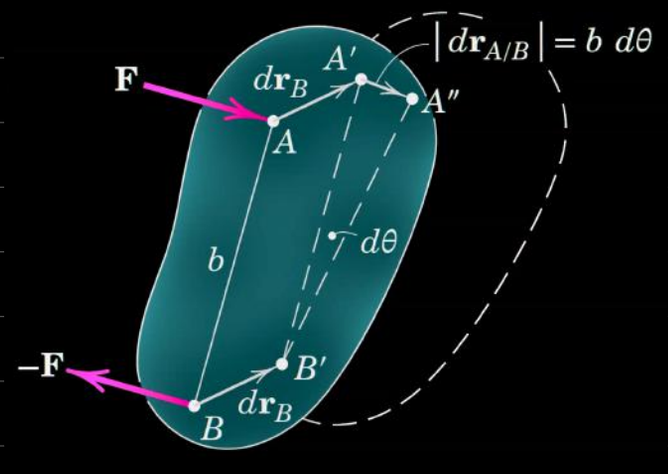
some arbitrary point O. Then, net torque about O:

$$\vec{M} = \vec{r}_1 \times \vec{F} + \vec{r}_2 \times (-\vec{F}) = (\vec{r}_1 - \vec{r}_2) \times \vec{F} = \vec{r}_{12} \times \vec{F}$$

now $\vec{r}_{12}$ is simply position vector of ① wrt ② which is independent of choice of point O. Thus net torque of a couple is same about every point OED.

→ Torque is also called moment, hence the symbol $\vec{M}$.

**Work Done By A Couple (Planar):** As a rigid body rotates under action of some couple, net work done by the couple is independent of choice of any translating frame. This can be easily shown as follows: Let $\vec{s}_1$ and $\vec{s}_2$ be the displacements of points on which forces are acting then net work done by the couple is $\vec{F}\cdot\vec{s}_1 - \vec{F}\cdot\vec{s}_2 = \vec{F}\cdot(\vec{s}_1 - \vec{s}_2)$ now $\vec{s}_1 - \vec{s}_2$ is displacement of particle 1 as seen from particle 2, and hence frame independent. Thus work done by a couple is frame independent. Now consider a rigid body with two points A and B and let F and −F be the components of forces acting on them ⊥ to line AB. Now if the rigid body turns through an angle dθ, displacement of A from frame of B is along an arc of length $b\,d\theta$. Thus $dW = Fb\,d\theta$ also $M = Fb$, thus $\boxed{dW = M\,d\theta}$

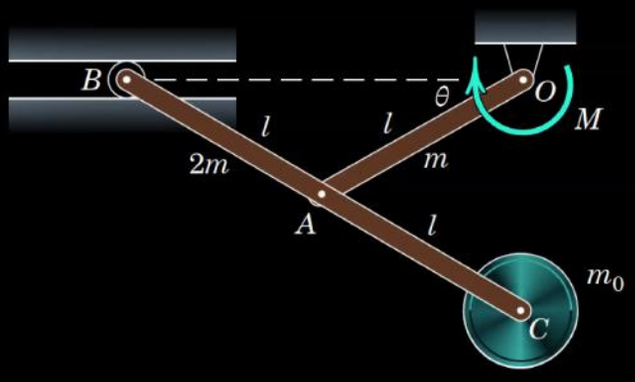



→ In case of 3D rotation, $dW = \vec{M}\cdot\vec{d\theta}$.

→ If rotation is in the sense of the couple, its work done is positive and if it is opposite to the sense of the couple, its work done is negative.

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**EXAMPLE: MERIAM KRAIGE:** Determine the couple that must be applied at O to keep the system in equilibrium at $\theta = 30°$. Masses of the disc and rods are shown in the figure. There is no friction.




**Solution:** Let us first write depth of CM of each massive member from point O.

For $m_0$ it is $2\ell \sin\theta$

For $m$ it is $\ell/2 \sin\theta$

and for $2m$ it is $\ell \sin\theta$, thus PE of the system: $-\ell \sin\theta \left[2m_0 + \frac{m}{2} + 2m\right]g$

i.e $U = -\frac{\ell}{2}\sin\theta\,[4m_0 + 5m]g$, now for small increase in angle $\theta$, work done by M must equal change in PE of the system. Thus:

$$-M\,d\theta = -\frac{\ell}{2}\cos\theta\, d\theta\,[4m_0 + 5m]g \Rightarrow M = \frac{(4m_0 + 5m)g\ell}{2}\cos\theta \text{ Ans.}$$

For $\theta = 30°$, $M = \frac{\sqrt{3}}{4}(4m_0 + 5m)g\ell$ Ans.

**POWER:** As defined earlier, Power is the rate of work done by any force, thus $P = dW/dt = \vec{F}\cdot d\vec{s}/dt = \vec{F}\cdot\vec{v}$ $\boxed{P = \vec{F}\cdot\vec{v}}$

Average power is defined as ratio of total work done to total time taken.

**POWER FORM OF WORK ENERGY THEOREM:** Power developed by all forces acting on a system is equal to rate of change of its kinetic Energy.

**EXAMPLE:**

The engine of a car of mass $m$ supplies a constant power P to the wheel to accelerate the car. Neglect ~~friction~~ (→ frictional losses) and air resistance. The car is initially at rest. Find the speed of car as a function of time. Also find displacement at a general time

(A) $(2Pt/m)^{1/2}$ (B) $2Pt/m$ (C) $(3Pt/m)^{1/2}$ (D) $(4Pt/m)^{1/2}$

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**Solution:** $P = dKE/dt = mv\,dv/dt \Rightarrow Pt = \frac{1}{2}mv^2$

(We could say this directly also). Now $v = \sqrt{\frac{2Pt}{m}} = \frac{ds}{dt} \Rightarrow s = \sqrt{\frac{2P}{m}} \cdot \frac{2}{3} t^{3/2}$

$\Rightarrow s = \sqrt{\frac{8P}{9m}}\, t^{3/2}$ Ans.

**EXAMPLE :**

A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration $a_c$ is varying with time as $a_c = k^2rt^2$, where k is a constant. The power delivered to the particle by the forces acting on it is:

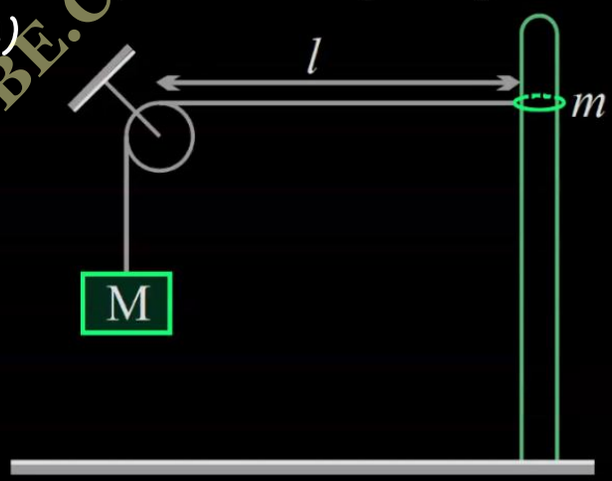
(A) $2\pi mk^2r^2t$ (B) $mk^2r^2t$ (C) $mk^4r^2t^5/3$ (D) 0

**Solution:** $ma_c = \frac{mv^2}{r} = mk^2rt^2 \Rightarrow KE = \frac{1}{2}mv^2 = \frac{1}{2}mk^2r^2t^2$

$\Rightarrow P = dKE/dt = mk^2r^2t$ Ans. (B).

**EXAMPLE :**

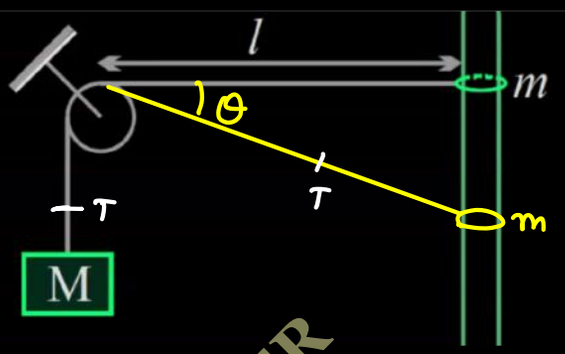
For the system shown in the figure, the mass m is initially held at rest. If now m is released from rest, locate its positions where power developed on the system by gravity is zero. (Find the angular positions of m wrt horizontal from the pulley) $(m < M)$




**Solution:** If power developed by gravity is zero, it means rate of change of KE of the system is zero (which also means rate of change of PE of the system is zero). Thus we wish to locate points of maximum and minimum KE (or PE). Clearly one such point is initial configuration itself i.e $\theta = 0°$.

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Now Max KE will occur at equilibrium position (equilibrium position means the position where system will stay in equilibrium if released from there, thus the position of force balance assuming zero KE).




At this position, $T = Mg = mg/\sin\theta$

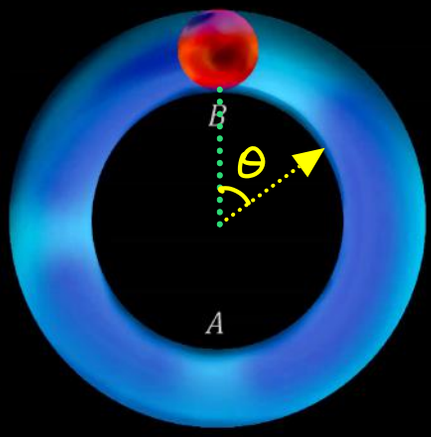
$\Rightarrow \sin\theta = m/M$ Ans.

Now min KE occurs at the lower extreme position of the ring. Here loss in gravitational PE of the ring is equal to gain in gravitational PE of the block. Thus:

$$mg\,l\tan\theta = Mg\,l(\sec\theta - 1) \Rightarrow \frac{\tan\theta}{\sec\theta - 1} = \frac{M}{m} = \frac{\sin\theta}{1-\cos\theta}$$

$\Rightarrow \cot(\theta/2) = M/m \Rightarrow \theta = 2\cot^{-1}(M/m)$ Ans.

**EXAMPLE:** A ball is released in a smooth vertical thin tube as shown. At what angular position is the power developed by gravity maximum?




**Solution:** For maximum power due to gravity, vertical velocity of the particle should be maximum, or vertical acceleration should be zero. Now at $\theta$ from vertical: $\frac{1}{2}mv^2 = mgR(1-\cos\theta)$ (1)

also $\frac{mv^2}{R} + N = mg\cos\theta$ (2), For zero vertical acceleration,

$N\cos\theta = mg$ (3) From (1), (2) & (3):

$$2mg(1-\cos\theta) + \frac{mg}{\cos\theta} = mg\cos\theta \Rightarrow \cos^2\theta = 2\cos\theta - 2\cos^2\theta + 1$$

$\Rightarrow 3\cos^2\theta - 2\cos\theta - 1 = 0 \Rightarrow \cos\theta = -\frac{1}{3}$ or $1$ but $1$ corresponds to zero power thus $\theta = \cos^{-1}\left(\frac{1}{3}\right)$ Ans.

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## EXAMPLE: PATHFINDER: CHECK-13:

A light elastic cord of length $l_0 = 5.0$ m and stiffness $k = 50$ N/m is inside a fixed box with a small portion of length $x_0 = 20$ cm left out of a hole on a wall of the box as shown in the figure. If the cord is pulled out of the box, a frictional force $F = 25$ N acts on the cord. How much work has to be done in slowly pulling out the entire length of the cord?



**Solution:** To pull the cord slowly, initially a gradually increasing force has to be applied until slipping starts (when pulling force becomes F). Till this time, the part outside the box extends by a distance $\Delta x = \frac{F x_0}{k l_0}$ and work done in the process $W_1 = \frac{1}{2} F \Delta x = \frac{1}{2} \frac{F^2 x_0}{k l_0}$ Now after this external force becomes constant and final length just before left end exits, $l = l_0 + F/k$ thus distance moved with constant force,

$S = l_0 + \frac{F}{k} - (x_0 + \Delta x) \Rightarrow W_2 = FS = F(l_0 - x_0) + \frac{F^2}{k} - F\Delta x$

$W_2 = F(l_0 - x_0) + \frac{F^2}{k} - \frac{F^2 x_0}{k l_0}$, Net work $W = W_1 + W_2$

thus $W = F(l_0 - x_0) + \frac{F^2}{k} - \frac{1}{2}\frac{F^2 x_0}{k l_0} = F(l_0 - x_0) + \frac{F^2}{k}\left(1 - \frac{x_0}{2l_0}\right)$

$= 132.25$ Ans.

**THINK:** How will you find heat liberated in the process?

**Solution:** Heat liberated will be the difference of work done by external agent and the change in stored potential energy in the spring.

## EXAMPLE: PATHFINDER: CHECK II:

A light elastic cord of force constant $k$ and relaxed length $l$ is suspended from the ceiling. A bead of mass $m$ threaded on the cord is held motionless close to the ceiling and released. If the bead slides down the cord with a constant speed $v$ relative to the ceiling, find expression for thermal power dissipated during sliding of the bead.




**Solution:** Thermal power developed must be equal to the difference between power developed by gravity and the rate of storage of

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spring potential energy. Now the distance travelled by the head at a time $t$ : $l = vt$, at this instant, let natural length of portion above head be $l_0$, then,

$$k\frac{l}{l_0}(vt - l_0) = mg \quad (1) \qquad (\because \text{force balance})$$

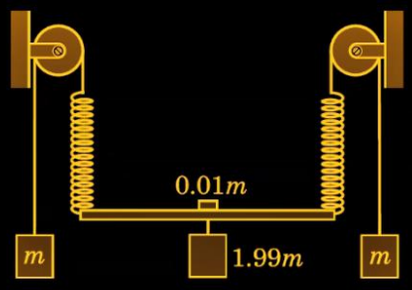
also Energy stored in string at this instant, $U = \frac{1}{2}\frac{kl}{l_0}(vt - l_0)^2$ (2)

From (1) & (2), $U = \frac{m^2g^2vt}{2(mg + kl)} \Rightarrow \frac{dU}{dt} = \frac{m^2g^2v}{2(mg + kl)}$

Now Rate of heat liberation: $\dot{Q} = mgv - \frac{dU}{dt} = \frac{mgv}{2}\left(\frac{mg + 2kl}{mg + kl}\right)$ Ans.

## EXAMPLE : PATHFINDER : BUILD-17 :

In the setup shown, an almost inertia-less bar is suspended horizontally with the help of two identical springs each of stiffness $k$, two light inextensible cords that pass over fixed ideal pulleys and two counterweights each of mass $m$. A small disc of mass $0.01m$ is placed at the midpoint of the bar. A block of mass $1.99m$ is suspended from the disc with the help of a light cord that passes through a hole in the bar. If this cord is cut, up to what maximum height will the disc jump?




Solution: Here we recognize that since $0.01m << m$, spring will contract suddenly with lower end having a much higher acceleration than the upper end. Thus practically entire spring energy will go in to imparting KE to the $0.01m$ (contact will be lost at almost natural length (so long as spring is applying upward force more than $0.01mg$, pseudo force and gravity both are downwards, hence ensuring presence of normal reaction). Now spring energy $= \frac{1}{2}Fx = \frac{F^2}{2k}$ Let disc jump through a height $h$ above the initial level then by conservation of mechanical energy :

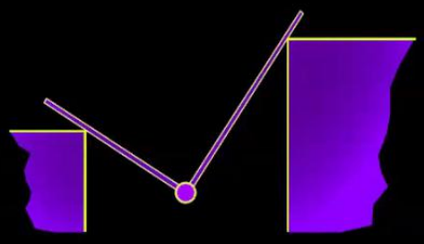
$$0.01mgh = \frac{(mg)^2}{2k} \times 2 \Rightarrow h = \frac{100mg}{k} \text{ Ans.}$$

→ A more exact analysis of this problem can be done using two body oscillation theory which we will learn in chapter on Simple Harmonic Motion.

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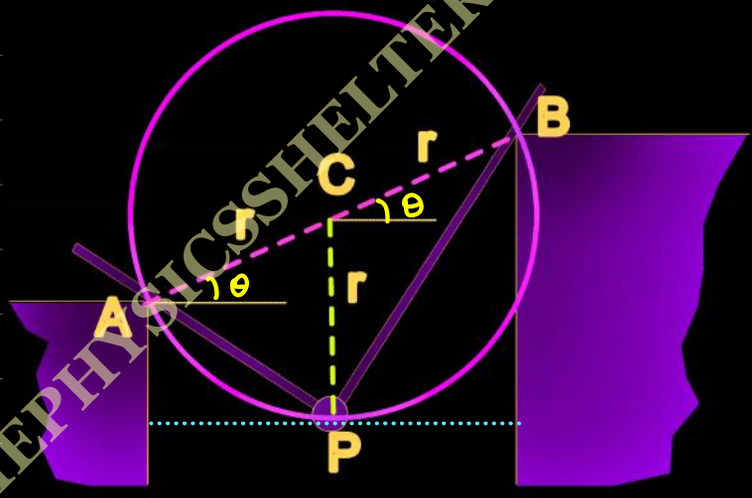
## Example: Pathfinder: MCQ: 17:

A light thin rod is bent into a right angle and a massive bead is affixed at the bend. The structure thus formed is placed on two platforms of different heights as shown in the figure. When the structure stays in equilibrium, what can you conclude regarding position of the ball relative to the vertical faces of the platforms?



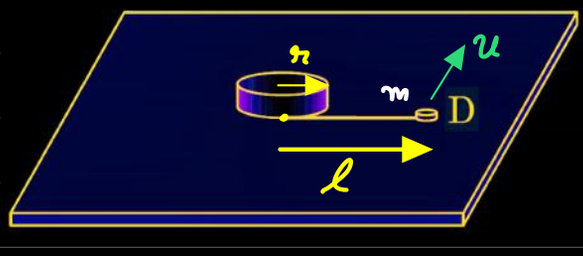
(a) The ball is closer to the lower platform than the higher one.
(b) The ball is closer to the higher platform than the lower one.
(c) The ball is equidistant from both the platforms irrespective of their heights.
(d) The ball is equidistant from both the platforms only when both the platforms are of the same height.

**Solution:** Consider the adjacent figure. Since angle APB is 90° and points A and B are fixed, we can imagine angle APB to be inscribed in a semi-circle with AB as diameter. P must be the lowest point of this circle to ensure least potential energy. Now as clear from figure, horizontal distance of the bead from both platforms is $r\cos\theta$ each. Thus bead is horizontally equidistant from both platforms (option c).




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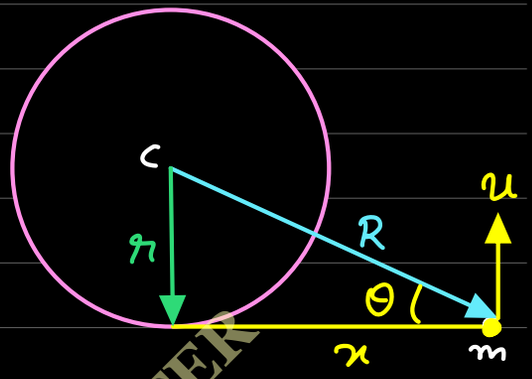
**Example:** On a frictionless plane, a string is wrapped around a fixed disc and the other end is connected to a point mass $m$ as shown. Point mass is flicked $\perp$ to the string so that the string starts wrapping around the fixed disc (a) What is the tension in the string at the initial moment (b) What is the power developed by tension (c) After how much time does the point mass collide with the disc.




**Solution:** The point of tangency on the disc is instantaneous radius of curvature of the spiral that disc will execute. Thus using centripetal force equation, $T = mu^2/\ell$ Ans. (a).

(b) Since tension always acts ⊥ to velocity its power developed is zero on the disc ∴ the speed of the disc remains constant.
(Tension will keep increasing as radius of curvature keeps decreasing due to wrappings of the string).

(c) Now consider a moment when unwrapped length is $x$. The approach velocity of the mass towards centre of the disc is $u\sin\theta$




i.e $$\frac{ur}{\sqrt{r^2+x^2}} = -\frac{dR}{dt} = -\frac{d}{dt}\left(\sqrt{r^2+x^2}\right)$$

$$\Rightarrow \frac{ur}{\sqrt{r^2+x^2}} = -\frac{1\cdot 2x\,dx/dt}{2\sqrt{r^2+x^2}} \quad \text{or} \quad x\frac{dx}{dt} = -ur \quad (1)$$

(We could also develop (1) directly, since $\omega = u/r$, unwrapped length in a time $dt$, $-dx = \frac{u}{r}dt \times r$ ($\because r\omega dt$).).

Now integrating (1) $$\int_0^l x\,dx = \int_0^T -ur\,dt \Rightarrow T = \frac{l^2}{2ur}$$ Ans.

→ Think: What is the length of path travelled until collision?

Solution: Path length $s = uT = l^2/2r$ Ans.

→ Think: What if there were viscous drag / kinetic friction

Solution: In that case, the frictional forces will act opposite to $u$
(in first case $dv/dt = -kv \Rightarrow v = ue^{-kt}$, in second case $v = u - \mu gt$)
now this value of $v$ can be substituted for $u$ in RHS of (1) and we can integrate again to get the desired results.

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